

Neha Verma

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Education

- **Johns Hopkins University** **Baltimore, MD**
Ph.D. in Computer Science, Advisors: Kevin Duh & Kenton Murray 2021 – 2026 (Expected)
 - **Johns Hopkins University** **Baltimore, MD**
M.S.E. in Computer Science 2021 – 2023
 - **Yale University** **New Haven, CT**
B.S. in Computer Science & Mathematics 2017–2021
– Distinction in the Major, Magna Cum Laude
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Experience

- **Google DeepMind & YouTube** **Mountain View, CA**
Research Scientist Intern 05/2025 – 08/2025
Developed a novel model-merging method for continual learning [2]. U.S. Patent Pending.
 - **Meta – Fundamental AI Research** **Menlo Park, CA**
Research Scientist Intern 05/2023 – 09/2023
Introduced a novel model-merging technique for Transformers that significantly reduces loss barriers between models trained from separate initializations [5].
 - **Johns Hopkins University** **Baltimore, MD**
Research Assistant 08/2021 – Present
Research toward my PhD. Topics include model merging for LLM compression [3, 4], representation learning for multilingual translation [1, 6, 7, 10], speech translation [8, 9], and domain adaptation [11].
 - **Language and Information Learning Lab (LILY) at Yale** **New Haven, CT**
Research Assistant 05/2018 – 08/2021
Undergraduate Research Assistant with Dragomir Radev. Researched structured text generation [13, 14], cross-lingual information retrieval [15], and NLP for EHRs [12].
 - **CVS Health/Aetna** **Remote**
Data Science Intern 05/2020 – 08/2020
Built a model to predict adverse hyperkalemic episodes for patients with kidney disease.
 - **Summer Undergrad Math Research at Yale (SUMRY)** **New Haven, CT**
Research Assistant – Advisor: Karamatou Yacoubou Djima 05/2019 – 08/2019
Researched spectral and harmonic methods for denoising medical vascular network images.
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Publications

- [1] N. Bafna, N. Verma, V. Zouhar, P. Koehn, D. Yarowsky. “There is no Theoretical Curse of Multilinguality for Embedding Space Structure”, *preprint*, 2026.
- [2] N. Verma*, N. Mehta*, S. Wang, N. Zhang, A. Tsai, A. Nath, L. Wei, L. Heldt, L. Hong, E. Chi, X. Yi. “ORBIT: Preserving Foundational Language Capabilities in GenRetrieval via Origin-Regulated

Merging”, *preprint*, 2026.

- [3] **N. Verma**, K. Murray, K. Duh. “DOTResize: Reducing LLM Width via Discrete Optimal Transport-based Neuron Merging”, *Weight-Space Symmetries: from Foundations to Practical Applications @ ICML 2026*.
- [4] **N. Verma**, K. Murray, K. Duh. “Merging feed-forward sublayers for compressed Transformers”, *TMLR 2026*.
- [5] **N. Verma**, M. Elbayad. “Merging text transformer models from different initializations”, *TMLR 2024 & High-dimensional Learning Dynamics (HiLD) @ ICML 2024*.
- [6] **N. Verma**, K. Murray, K. Duh. “Exploring geometric representational disparities between multilingual and bilingual translation models”, *LREC-COLING 2024*.
- [7] E. Salesky, **N. Verma**, P. Koehn, M. Post. “Multilingual Pixel Representations for Translation and Effective Cross-lingual Transfer”, *EMNLP*, 2023.
- [8] H.L. Xinyuan*, **N. Verma***, B. Odoom, U. Pradeep, M. Wiesner, S. Khudanpur. “JHU IWSLT 2023 Multilingual Speech Translation System Description”, *IWSLT*, 2023.
- [9] A. Hussein, C. Xiao, **N. Verma**, T. Thebaud, M. Wiesner, S. Khudanpur. “JHU IWSLT 2023 Dialect Speech Translation System Description”, *IWSLT*, 2023.
- [10] K. Marchisio, **N. Verma**, K. Duh, P. Koehn. “IsoVec: Controlling the Relative Isomorphism of Word Embedding Spaces”, *EMNLP*, 2022.
- [11] **N. Verma**, K. Murray, K. Duh. “Strategies for Adapting Multilingual Pre-training for Domain-Specific Machine Translation”, *AMTA*, 2022.
- [12] I. Li, J. Pan, J. Goldwasser, **N. Verma**, W. P. Wong, M. Nuzumlalı, B. Rosand, Y. Li, M. Zhang, D. Chang, R. A. Taylor, H. M. Krumholz, D. Radev. “Neural Natural Language Processing for Unstructured Data in Electronic Health Records: a Review”, *Computer Science Review*, 2022.
- [13] L. Nan, C. Hsieh, Z. Mao, X. Lin, **N. Verma**, R. Zhang, W. Kryściński, N. Schoelkopf, R. Kong, X. Tang, M. Mutuma, B. Rosand, I. Trindade, R. Bandaru, J. Cunningham, C. Xiong, D. Radev. “FeTaQA: Free-form Table Question Answering”, *TACL*, 2021.
- [14] L. Nan, D. Radev, R. Zhang, A. Rau, A. Sivaprasad, C. Hsieh, X. Tang, A. Vyas, **N. Verma**, P. Krishna, Y. Liu. “DART: Open-domain structured data record to text generation”, *NAACL*, 2021.
- [15] R. Zhang, C. Westerfield, S. Shim, G. Bingham, A. Fabbri, W. Hu, **N. Verma**, and D. Radev. “Improving Low-Resource Cross-lingual document Retrieval by Reranking with Deep Bilingual Representations”, *ACL*, 2019.

Awards and Grants

- Phi Beta Kappa 2021
 - Yale College First-Year Summer Research Fellowship in the Sciences & Engineering 2018
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Teaching

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| • Introduction to Human Language Technology
<i>Teaching Assistant</i> | Johns Hopkins University
<i>Fall 2025</i> |
| • Language and Computation
<i>Course Assistant</i> | Yale University
<i>Spring 2021</i> |
| • Fields and Galois Theory
<i>Peer Tutor</i> | Yale University
<i>Spring 2020, Spring 2021</i> |
| • Abstract Algebra
<i>Course Assistant</i> | Yale University
<i>Fall 2019, Fall 2020</i> |
| • Linear Algebra & Vector Calculus II
<i>Course Assistant</i> | Yale University
<i>Spring 2019</i> |
| • Calculus of Functions of Several Variables
<i>Peer Tutor</i> | Yale University
<i>Fall 2018</i> |

Service and Professional Activities

Organizing

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| • JHU CLSP Application Support Program | <i>2022-2025</i> |
| • Machine Translation Marathon of the Americas (MTMA) | <i>2024</i> |
| • Regional NACLO @ Yale University | <i>2020-2021</i> |
| • Yale Association for Women in Mathematics (AWM) Chapter President | <i>2020-2021</i> |

Reviewing

- **Conferences:** COLM, ICML, ICLR, NeurIPS, ARR, EMNLP, ACL, and AMTA
- **Journal:** Computational Linguistics (2022)

Skills

- **Programming Languages:** Python, Bash, SQL
- **Frameworks/Toolkits:** PyTorch, Jax, fairseq, Hugging Face, Dask, $\text{\LaTeX}$, Git